

No. of Functional Features Included in the Solution

Date	30 June 2026
Team ID	
Project Name	A comprehensive measure of well being
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Maximum Marks	5 Marks

Functional Features Overview:

This document lists the functional features included in the HDI Predictor solution, each described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Data Preprocessing	Cleans and prepares life expectancy, schooling, and GNI data from UNDP/Kaggle sources for model training	Backend / Data	Done	1
2	HDI Classification Model	Trained scikit-learn model that predicts a country's HDI tier (Very High/High/Medium/Low) from its indicators	Backend / ML	Done	1.5
3	Prediction Web Form	Flask form where users enter life expectancy, schooling years, and GNI per capita	UI / Frontend	Done	1
4	Prediction Result Display	Shows the predicted HDI tier and a short explanation immediately after form submission	UI / Frontend	Done	1
5	Indicator Visualization	Matplotlib/Seaborn charts showing which indicators most influenced the prediction	UI / Backend	In Progress	0.5
6	Scenario Coverage	Supports and validates the three project scenarios: high-performing, mid-range, and low-development countries	Backend / ML	Done	1
7	Model Persistence	Saves and loads the trained model as a .pkl file for reuse without retraining	Backend	Done	0.5
8	Input Validation	Validates that entered indicator values are within realistic ranges before prediction	Backend / Frontend	In Progress	0.5

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	6
Core / Must-Have Features	5 (Preprocessing, Classification Model, Web Form, Result Display, Scenario Coverage)
Additional / Nice-to-Have Features	3 (Indicator Visualization, Model Persistence, Input Validation)
Features Tested & Verified	6

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
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1	User Interface (UI)	2	Prediction web form, prediction result display
2	Backend / Logic	3	Data preprocessing, model persistence, input validation
3	Database / Storage	1	UNDP/Kaggle HDI dataset storage and pickled model file
4	API / Integration	1	Flask routes connecting the web form to the ML prediction engine
5	Security / Authentication	0	Not required for current MVP scope (open-access prediction tool)